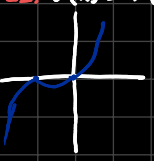
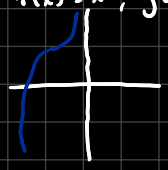
1. $f(x) = 4 - x - 3x^2$, degree: 2, leading term: $-3x^2$, constant term: 4, end behavior: $x \to -\infty, f(x) \to -\infty$; $x \to \infty, f(x) \to -\infty$

11. $a(x) = x(x+2)^2$, z: $x = 0$, $x = -2$



21. $f(x) = x^3$, $g(x) = (x+2)^3 + 1$; D: $(-\infty, \infty)$ R: $(-\infty, \infty)$

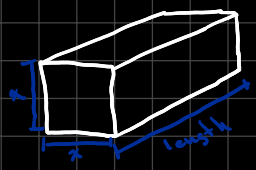


$(0,0) \to (-2,1)$
$(1,1) \to (-1,2)$
$(-1,-1) \to (-3,0)$

27. $f(x) = x^3 - 9x + 5$; $[-4,-3]$, $[0,1]$, $[2,3]$

$\left.\begin{matrix} f(-4) = -23 \\ f(-3) = 5 \end{matrix}\right\}$ sign change $\quad \left.\begin{matrix} f(0) = 5 \\ f(1) = -3 \end{matrix}\right\}$ sign change $\quad \left.\begin{matrix} f(2) = -5 \\ f(3) = 5 \end{matrix}\right\}$ sign change

33. $\underbrace{\text{Length} + \text{Girth} \leq 130}_{\text{Parcel Post}}$ $\quad \underbrace{\text{Length} + \text{Girth} \leq 108}_{\text{Other}}$



$G(x) = 4x$

a) $L + G = 130 \to L = 130 - G(x) = 130 - 4x$
$V = (L)(x)(x) = Lx^2 = (130 - 4x)x^2 = 130x^2 - 4x^3$

b) $V_{max} = (21.66\ldots, 20342.59)$

c) $L + G = 108$, $L = 108 - G(x) = 108 - 4x$; $V(x) = L(x)(x)(x) = (108 - 4x)x^2 = 108x^2 - 4x^3$

$V_{max} = (18, 11664)$